

Dhruv Shah

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EDUCATION

Carnegie Mellon University

GPA 3.73, August 2025 – May 2029

B.S. in Electrical and Computer Engineering, Minor in AI

Pittsburgh, PA

Relevant Coursework: Digital Systems, Data Structures & Algorithms, Signals and Systems, Discrete Mathematics, Probability

TECHNICAL EXPERIENCE

AI/ML Intern | Netramind Innovations

May 2026 – August 2026

- Developed a deep learning pipeline for ultra-widefield fluorescein angiography segmentation of retinal total area and non-perfusion area, benchmarking UNet, UNet++, MANet, MAE-pretrained, dual-decoder, and CNN-RNN variants; best model achieved 0.963 TA Dice, 0.494 NPA Dice, and 0.879 NPA AUC on a patient-level held-out test set.
- Designed temporal FA sequence experiments using 624 multiframe images across 70 cases, implementing QC-filtered frame selection, probability-based mask collation, and CNN-RNN architectures with LSTM/ConvGRU fusion to study whether angiography time-series context improves NPA segmentation.

Software Engineering Intern | Lockheed Martin

May 2025 – July 2025

- Developed a Python wrapper for Stable Diffusion XL (SDXL), integrating REST API endpoints and JSON-based inference requests to expose configurable generation parameters including resolution, guidance scale, seed, and batch size.
- Integrated SDXL into internal generative-AI workflows for automated image and logo generation, while prototyping LangChain-based LLM orchestration and agentic workflows for multimodal AI applications.

Computational Research Assistant | University of Florida

May 2024 – July 2024

- Adapted a FORTRAN-based compressible-flow CFD solver (HyBurn) to simulate conventional, supercritical, and diamond airfoils using immersed-boundary .STL geometries across Mach 0.5, 0.8, 1.2, and 1.5.
- Developed Python-based analysis workflows to process CFD outputs and evaluate lift/drag coefficients, pressure distributions, and shock-wave behavior across subsonic, transonic, and supersonic flow regimes.
- Compared aerodynamic performance across airfoil geometries and operating conditions to characterize Mach-dependent flow behavior; synthesized results into a 4,000-word research paper, technical poster, and formal presentation.

PROJECTS

FPGA Combinational Logic Design | SystemVerilog, Vivado, Boolean Board

- Designed/implemented a 6-input, 2-output combinational logic system in structural SystemVerilog, deriving minimized SOP/POS representations via 64-entry truth tables and Karnaugh maps and mapping designs to constrained NAND/NOR gate-level architectures.
- Developed SystemVerilog testbenches for functional verification, synthesized and deployed the design to a Boolean Board FPGA using Vivado, and compared alternative implementations through gate count and synthesized Slice LUT utilization.

Embedded Sensor Interface | Raspberry Pi Pico, I²C, CircuitPython

- Interfaced a Raspberry Pi Pico with a TMP102 temperature sensor, LCD, and button/LED peripherals using I²C and GPIO, implementing register-level sensor configuration and real-time data acquisition.
- Debugged hardware/software communication using an Analog Discovery 3 logic analyzer, inspecting I²C transactions and validating peripheral addressing, register reads/writes, and sensor data.

LEADERSHIP

Orientation Counselor | Carnegie Mellon University

Aug. 2026

- Selected to mentor incoming students during CMU Orientation, leading group of 30 students through programming and coordinating with university staff to support first-year transition.

AWARDS

Dean's List | SAT: 1550/1600 | ACT: 35/36 | 7th Nationally, BPA Economic Research | 13th Nationally, Science Olympiad Elastic-Launched Glider | 11th Statewide, Mu Alpha Theta Calculus Division | Presidential Award for Academic Excellence | National Merit Finalist

SKILLS

Languages: C/C++, SystemVerilog, Python, Java, FORTRAN, JavaScript, Swift

Digital Design: FPGA, RTL Design, Digital Logic, Combinational/Sequential Logic, Logic Synthesis, SystemVerilog Testbenches

Embedded Systems: Raspberry Pi, Raspberry Pi Pico, Microcontrollers, I²C, GPIO, Sensors & Peripheral Interfacing

Tools & Software: Vivado, Analog Discovery 3, Logic Analyzer, Git/GitHub, Linux/Unix, Jupyter, REST APIs

AI/ML: PyTorch, TensorFlow/Keras, OpenCV, NumPy, Computer Vision

Languages: English (fluent/native), Spanish (fluent)